International Islamic University Chittagong
Department of Computer Science and Engineering (CSE)
B.Sc. Engineering in CSE
Final Examination, Spring 2025

Course Code: EEE-2421 **Course Title:** Electrical Drives and Instrumentation
Time: 2.5 hours **Full Marks:** 50

Answer all the questions. The figures in the right-hand margin indicate full marks
Course Outcome (COs) and Bloom's Level (BL) are mentioned in additional Columns

Group-A

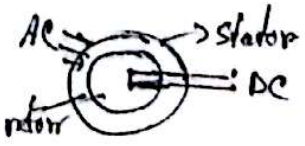
1 (a)	Explain the working principle of a Variable Frequency Drive using a block diagram.	CO2	C	5
1 (b)	A 4-pole, 3-phase induction motor operates from a supply with a frequency of 50 Hz. Calculate: i. The speed at which the magnetic field of the stator is rotating. ii. The speed of the motor when the slip is 3%. iii. The speed of the rotor when the slip is 0.04. iv. The frequency of the rotor currents when the slip is 0.03. v. The frequency of the rotor currents at standstill.	CO3	E	5
	Or			
1 (a)	Briefly explain the working principle of three phase induction motor. Discuss how Variable Frequency Drive (VFD) controls the speed of an induction motor.	CO1	U	5
1 (b)	A 3-φ induction motor is wound for 5 poles and is supplied from 50-Hz system. Calculate (i) the synchronous speed (ii) the rotor speed when slip is 5% and (iii) rotor frequency when rotor runs at 500 rpm.	CO1	A	5
2 (a)	Suppose you are working in an industry. You need a motor with constant speed and that can operate in lagging to leading power factor for some applications. Select an appropriate motor for your desired application, explain the working principle of your selected motor, and the reasons for your selection.	CO2	C	5
2 (b)	Briefly analyze the operation of the variable reluctance stepper motor and the permanent magnet stepper motor, and compare them.	CO3	An	5

Group-B

3 (a)	Discuss the three stages of the generalized measurement system and types of errors in the measurement system.	CO3	U	5
3 (b)	For the following given data, $X_1 = 43.7$, $X_2 = 49.1$, $X_3 = 51.2$, $X_4 = 46.6$, $X_5 = 44.7$, Calculate: a) Arithmetic mean b) Deviation of each value c) Algebraic sum of the deviations d) The average deviation e) The standard deviation	CO3	E	5

$N_s = \frac{120 f_s}{P}$

$f_r = s f_s$



$E = C(T_2 - T_1) + B(T_2^2 - T_1^2)$

Or

3 (a)	Define the terms Accuracy and Precision. Also, describe the complete process of measurement by explaining the steps taken **before, during,** and **after** the measurement.	CO2	R	5
3 (b)	The following 10 voltage readings were recorded during an experiment: **54.58, 54.73, 54.86, 54.95, 55.00, 55.00, 55.08, 55.19, 55.31, 55.30** volts. Calculate the following: i) Arithmetic Mean ii) Deviation of each value iii) Average Deviation iv) Standard Deviation v) Variance	CO2	An	5
4 (a)	Sketch the three-op-amp instrumentation amplifier and derive the expression for the overall gain. Also, mention the advantages of using this configuration.	CO2	Ap	5
4 (b)	Explain with the help of a functional block diagram, the principle of operation of a digital frequency meter.	CO3	C	5
5 (a)	I. A displacement transducer with a shaft stroke of 3 inch is applied to the circuit of Figure 1. The total resistance of the potentiometer is 5kΩ, and the applied voltage V_T = 5V. When the wiper is 0.9 inch from B, what is the value of the output voltage V_O?	CO3	E	5

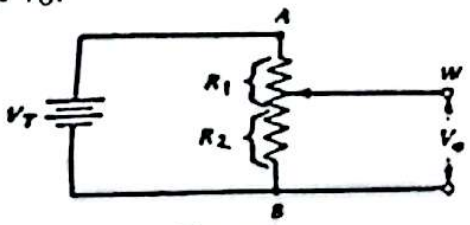


Figure 1.

II. During experiments with a copper-constantan thermocouple, it was found that $c = 3.75 \times 10^{-2}$ mV/°C and $k = 4.5 \times 10^{-5}$ mV/°C^2. If T_1 = 1000 °C and the cold junction T_2 is kept in ice, compute the resultant electromotive force.

5 (b)	What is a transducer? Discuss the classification of transducers and prove the gauge factor equation for a strain gauge.	CO2	U	5

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International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering

Final Examination, Autumn 2024 Program: B. Sc. In CSE

Course Code: **EEE-2421** Course Title: **Electrical Drives and Instrumentation**

Time: 2 hour 30 minutes Full Marks: 50

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.

(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Part-A					
1)	a)	What is a synchronous motor? Describe its working principle and illustrate with an appropriate diagram.	CO2	U	5
1)	b)	An induction motor has a synchronous speed of 1500 RPM and a rotor speed of 1450 RPM. (i) Calculate the slip of the motor in percentage. (ii) What is the rotor speed of this motor at the rated load? (iii) What is the rotor frequency of this motor at the rated load?	CO3	An	5
2)	a)	Define slip and explain the different types of rotors used in stepper motors, along with appropriate diagrams.	CO3	An	5
2)	b)	Explain the concept of a stepper motor. Also, describe the full-step operation mode with a suitable diagram.	CO1	Ap	5
		OR			
2)	a)	A 4 pole, 3 phase induction motor operates from a supply whose frequency is 50Hz calculate: i) the Speed of the stator magnetic field ii) the Speed of rotor when slip is 0.04 iii) the Speed Frequency of the rotor currents when the slip is 0.03 the Speed Frequency of the rotor current at standstill.	CO3	An	5
2)	b)	How is stepper motor different than induction motor? Describe Full-Step mode steps of a stepper motor.	CO3	Ap	5

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		Part-B			
3)	a)	Define measurement error and classify its types. Provide examples of each type and explain how they can affect the accuracy of a measurement system.	CO2	An	5
		OR Identify the importance of measurement. Write short notes on the following topics: (i) Direct vs. Indirect Methods in Generalized Systems (ii) Absolute Instruments vs. Secondary Instruments			
3)	b)	Given two voltage measurements • $V1=100V\pm1\%$ • $V2=80V\pm5\%$ Calculate: i) The maximum percentage error for each of the voltage measurements, ii) The maximum percentage error in the difference between the two voltages.	CO3	An	5
		OR The resistance values of six precision resistors are measured by comparing each of them to a 1.000 Ω standard resistor. The measured resistances are: R1 = 1.003 Ω, R2 = 1.004 Ω, R3 = 1.001 Ω, R4 = 0.997 Ω, R5 = 0.999 Ω, and R6 = 1.002 Ω. Calculate: 1. The average measured resistance, 2. The average deviation, 3. The standard deviation, The probable measurement error.			
4)	a)	Sketch a circuit diagram of a logarithmic (log) amplifier and derive the equation for its output voltage.	CO4	U	5
4)	b)	Sketch a common anode 7-segment display and provide its truth table for the display logic.	CO4	Ap	5
5)	a)	Explain the concept of optoelectronics. Provide a brief overview of how a solar cell works and mention its applications.	CO3	An	5
5)	b)	What is a transducer? Derive the formula for gauge factor in case of a strain gauge. or During an experiment with a copper-constantan thermocouple, it was found that c = 3.75 X 10-2 mV/°C and k = 4.5 x 10-5 mV/°C. If T1 =100°C and the cold junction T2 is kept in ice. Compute the EMF.	CO3	Ap	5

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International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

B.Sc in CSE Final Examination, **Spring - 2024**

Course Code: EEE-2421 **Course Title:** Electrical Drives and Instrumentation

Full Marks: 50 **Time:** 2 hour 30 minutes

[Answer all the five questions. Figures in the right-hand margin indicate full marks]

Course outcomes and Bloom's Levels are mentioned in additional columns

	Part-A	**CLO**	**DL**	**Marks**
1(a)	"Sometimes Induction Motor is called as Rotating Transformer."-Justify the statement. Briefly Explain the working principle of the induction motor. Also draw a neat diagram of its Exact equivalent circuit.	CO2	U	1+4
1(b)	A 480-V, 50-Hz, 50-hp, three-phase induction motor is drawing 60 A at 0.85 PF lagging. The stator copper losses are 2.1 kW, and the rotor copper losses are 300 W. The friction and wind age losses are 500 W, the core losses are 1800 W, and the stray losses are negligible. Find the following quantities: **i.** The air-gap power **ii.** The power converted **iii.** The output power **iv.** The efficiency of the motor	CO3	An	5
	OR			
1(b)	A 208-V, 10-hp, four-pole, 60-Hz, Y- connected induction motor has a full-load slip of 5 percent i) What is the synchronous speed of this motor? ii) What is the rotor speed of this motor at the rated load? *iii)* What is the rotor frequency of this motor at the rated load? *iv) What is the shaft torque of this motor at the rated load?*	CO3	An	5
2(a)	*What do you understand by the term "Slip"? A 6-pole, 3-phase induction motor operates from a supply whose frequency is 60 Hz. Calculate:* **i.** *Magnetic field speed* **ii.** Rotor speed **iii.** Frequency of rotor current at standstill.	CO3	An	3
	OR			
2(a)	A hybrid VR stepping motor has 8 main poles which have been castellated to have 5 teeth each. If the rotor has 50 teeth, calculate- **i.** Step angle **ii.** Resolution	CO3	An	3
2(b)	How to make a synchronous motor self-starting? Explain.	CO2	U	2
	OR			
2(b)	Write different types of Induction motor rotor and describe reasons behind this type of naming.	CO2	U	2
2(c)	When a 3 – φ power supply is connected with the stator of a 3-φ Induction Motor then a rotating magnetic field creates and rotates around the stator. Explain how it happens? Also draw the transformer equivalent circuit of induction motor.	CO2	U	5

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Part-B

3(a)	What do you understand by the term measurement and instrumentation? Why measurement is so important in our life?	CO2	U	3
	OR			
3(a)	What is Measurement? Briefly explain the steps of generalized measuring system with proper example.	CO2	U	3
3(b)	Why Measurement accuracy and precision need in a measurement system?	CO2	U	2
3(c)	Suppose you have a voltage generator. You want to measure the frequency of the voltage signal using digital technique. But you do not have any device to measure frequency. So, you decided to design a frequency counter. Draw the block diagram of circuit and explain each block.	CO3	An	5
4(a)	The accuracy of five precision resistors is checked by comparing each of them to a 1.0 Ω standard resistor. The measured resistances are as follows: R1=1.001 Ω, R2=1.002 Ω, R3=0.999 Ω, R4=0.998 Ω and R5=1.000 Ω. Calculate: i. The average measured resistance ii. The average deviation iii. The standard deviation iv. The probable measurement error	CO3	An	5
4(b)	Explain Photo-Voltaic effect briefly.	CO2	U	3
	OR			
4(b)	Briefly explain the working principle of PV cell. Also mention advantages and disadvantages of PV cell.	CO2	U	3
4(c)	An op-amp has a common mode gain 5. If 20v and 16v signals are given to input terminal, then a 40v output has come. Write your comments on whether this is a good op-amp or not good op-amp based on your analysis.	CO3	An	5
5(a)	Briefly explain the working of Resistance Temperature Detector (RTD) and also write the applications of RTD.	CO4	U	5
	OR			
5(a)	What is transducer? Explain the working principle of Piezoelectric sensor with appropriate figure.	CO4	U	5
5(b)	Consider two voltage measurements V1 and V2 with the following specified values and tolerances: • $V1=100V \pm 1\%$ • $V2=80V \pm 5\%$ i. Calculate the maximum percentage error of two voltage measurements ii. Calculate the maximum percentage error in the difference of two voltage measured voltages	CO3	An	5

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Exam, Autumn 2023

Course Code: **CSE-2421** Course Title: **Electrical Drive and Instrumentation**

Time: 2 hours 30 minutes Full Marks: 50

		Group A			
1.	a)	What is slip? Explain the production of rotational magnetic field in induction motor with appropriate diagram.	CO1	Rm	3
1	b)	Draw the transformer equivalent circuit of 3-Phase Induction Motor.	CO1	Un	3
1.	c)	A 208V, 10-hp, four-pole, 50-Hz, Y-connected induction motor has a full-load slip of 5%. (i) What is the synchronous speed of this motor? (ii) What is the rotor speed of this motor at the rated load? (iii) What is the rotor frequency of this motor at the rated load? (iv) What is the shaft torque of this motor at the rated load?	CO1	An	4
2.	a)	Discuss the full step mode operation of the Stepper Motor with neat diagrams.	CO1	Un	5
2.	b)	Explain the working principle of Variable frequency drive with proper diagram.	CO3	Un	5
		OR			
2.	b)	What is the difference between Induction Motor and Synchronous Motor? Explain a method to make Synchronous Motor Self-Starting.	CO3	Un	5
		Group B			
3.	a)	What are the different types of errors that can occur in the process of measurement? Briefly explain systematic errors.	CO3	An	5
3.	b)	What do you understand by the term measurement and instrumentation. What steps comprise up the Generalized Measuring System? Also, provide an example for each step.	CO3	Ap	5
4.	a)	Explain the operation of Solar Cell with proper diagram.	CO4	Ap	5
4.	b)	Derive the output voltage equation of logarithmic amplifier with transistor and op-amp.	CO4	Ap	5
		OR			
4.	b)	What is Optoelectronics. Mention different types of optoelectronic devices and explain the basic principle of Light Emitting Diode (LED).	CO4	Un	5
5.	a)	Sketch 7-segment display with truth table for both common cathode mode and common anode mode.	CO4	Un	5
5.	b)	What is transducer? Prove the gauge factor equation for strain gauge.	CO4	Un	5

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. Engineering in CSE

Final Examination, Spring 2023

Course Code: EEE 2421 Course Title: **Electrical Drives and Instrumentation**

Time: 2 hour 30 minutes Full Marks: 50

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.

(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Part -A

1)	a)	Explain the key principles and techniques involved in the operation of an inverter, which coverts DC into AC power effectively and reliably.	C01	5
1)	b)	Explain the working principle of a Variable Frequency Drive using a block diagram.	C01	5
		Or		
1)	a)	"An induction motor can't run at synchronous speed"- verify the statement with proper mathematical expression.	C01	5
1)	b)	A three-phase induction motor has a synchronous speed of 1800 RPM and a slip of 4%. Determine the actual speed of the motor in RPM.	C01	3
1)	c)	Write a short note on the constructional features of an induction motor.	C01	2
2)	a)	Explain how does a synchronous motor operate, and what mechanism does it employ to remain in sync with the rotating magnetic field?	C01	5
2)	b)	Write how many driving techniques are there for stepper motor. Discuss the half-step mode steps operation of Stepper motor with diagrams.	C01	5

Part-B

3)	a)	Write down the necessity of measurement.	C03	3
3)	b)	What are the different types of errors that can occur in the process of measurement? Briefly explain systematic errors.	C03	5
3)	c)	Differentiate between the term accuracy and precision.	C03	2
4)	a)	Draw a neat diagram of a logarithmic amplifier and derive the equation $V_o = -\eta V_T \ln (V_i / I_o R)$	C04	5
4)	b)	Explain the operation of differential amplifier using BJT.	C04	5
		Or		
4)	a)	Explain the Biomedical application of a chopper amplifier with proper block diagram and brief description.	C04	4
4)	b)	What is CMRR? Provide proper equation.	C04	3
4)	c)	Sketch 7-segment display with its truth table.	C04	3
5)	a)	Explain the operation of Solar cell with proper diagram.	C04	4
5)	b)	In what ways are piezoelectric transducers are used in different industries and fields?	C04	3
5)	c)	Mention different types of optoelectronic devices and explain optical fiber.	C04	3

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International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Semester Final Exam, Spring 2022
Course Code: EEE-2421 Course Title: Electrical Drives and Instrumentations
Total Marks: 50 Time: 2.5 Hours
[Answer *all the five* questions. Figures in the right-hand margin indicate full marks.]

Course Outcomes and Bloom's Levels are mentioned in additional Columns

		PLO	CLO	Marks
	PART A			
1.(a)	What is induction motor and why is it called?	4	3 C2	2
(b)	Explain the operation of capacitor start induction motor with example.	4	3 C2	3
	OR			
	Briefly explain the working principle of the induction motor. Also draw a neat diagram of its exact equivalent circuit.			
(c)	Explain how induction motor rotor rotates.	4	3 C2	2
(d)	A 3-phase induction motor is wound for 4 poles and is supplied from 50Hz system. Calculate i. Synchronous speed, ii. The rotor speed, when slip is 4% and iii. Rotor frequency when runs at 600rpm.	4	3 C3	3
2.(a)	Explain the full step operation of stepper motor and discuss the significance of stepper motor in modern automation industries	4	3 C3	5
	OR			
	Describe Half wave mode Driving technique for Stepper motor with neat diagram.			
(b)	Why synchronous motor has no self-starting torque? Write some methodes to make SM Self-Starting.	4	3 C3	2.5
(c)	Suppose you are working in an industry. You need a motor with constanat speed and can operate in lagging to leading power factor for some specific applications. Select an appropriate motor for yor desired application and explain the reasons of your selection.	4	3 C3	2.5
	PART B			
3.(a)	Write the process of measuring system with an appropriate example (based on the lab project done by yourself).	2	4 C4	3
(b)	What do you understand by the term measurement and instrumentation? Why measurement is so important in our life?		4 C4	3
(c)	Explain the significance of Measurement error combinations. If following voltage measurements are considered from two instruments. $V_1 = 90 \pm 3\ \%$, $V_2 = 80 \pm 5\ \%$. Calculate a. the maximum percentage error of two voltage measurements. b. the maximum percentage error in the difference of two voltages	2	4 C4	4
	OR			
	Explain Common Anode and Common Cathode "7-segment Display" with necessary diagram.	2	4 C4	4

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4.(a)	Explain with the help of a functional block diagram, the principle of operation of a digital frequency meter	2	5 C4	5	
	OR				
	For the following given data, calculate I) Arithmetic mean, ii) Deviation of each value, iii) Algebraic sum of the deviations, iv) Calculate the average deviation and v) Calculate the standard deviation Given, Z1 = 70.7, Z2 = 71.1, Z3 = 70.2, Z4 = 69.6 and Z5 = 71.7	2	5 C4	5	
(b)	(i) What are the advantages of digital systems over analog? (ii) Explain the following terms as applied to digital measurement. (a) Resolution (b) Sensitivity of digital meter (c) Accuracy specification of digital meters	2	5 C4	5	
5(a)	Write a difference between thermocouple, RTD and Thermistor. Suppose you need to measure temperature from -190°C to 450°C, which device will be suitable for measuring.	3	5 C4	5	
(b)	During experiments with a copper-constantan thermocouple it was found that c = 3.75 x 10^{-2} mV/0C and k = 4.5 x 10^{-5} mV/0C2. If T_1 = 1000C and the cold junction T_2 is kept in ice, compute the resultant electromotive force.	1	5 C3	2	
	OR				
	How Electricity is produced in a PV Panel? Write the prospects of Solar energy in Bangladesh.	1	5 C3	2	
(c)	What is transducer? Why Transducer is important? Explain different optoelectronic transducer in brief.	1	5 C3	3	

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